

SARASWAT ACADEMY

Class Test - Chapter 3: Tissues in Action

Grade: 9 | Time: 1 Hour | Total Marks: 20

Instructions:

- Answer all questions.
- Write legibly and clearly.
- Show all calculations and reasoning.
- Questions are of varying difficulty levels.

Section A: Multiple Choice Questions (5 marks)

1. Which type of meristematic tissue is responsible for the growth of plants in length?

- (a) Lateral meristem (b) Apical meristem (c) Intercalary meristem
(d) Permanent tissue

2. The epidermis of plants is primarily responsible for:

- (a) Photosynthesis (b) Storage of food (c) Protection and transpiration
(d) Transportation of water

3. Which tissue in animals is responsible for conducting electrical impulses throughout the body?

- (a) Epithelial tissue (b) Connective tissue (c) Nervous tissue (d) Muscular tissue

4. Sclerenchyma cells have thickened walls due to deposition of:

(a) Cellulose (b) Lignin (c) Pectin (d) Chlorophyll

5. The hinge joint in the human body allows movement in:

(a) All directions (b) One direction only (c) Two directions (d) Three directions

Section B: Assertion-Reason Questions (2 marks)

Direction: In the question given below, there are two statements - Assertion (A) and Reason (R). Select the correct option:

- (i) Both A and R are true, and R is the correct explanation of A.
- (ii) Both A and R are true, but R is not the correct explanation of A.
- (iii) A is true, but R is false.
- (iv) A is false, but R is true.

Assertion (A): Meristematic tissues lack vacuoles and have densely packed cells.

Reason (R): The absence of vacuoles and dense cytoplasm allow continuous and rapid cell division in meristematic tissues.

Section C: Case Study Question (4 marks)

A gardener observed that when a young mango plant's stem tip was carefully cut, the plant stopped growing in length but soon developed new branches from the nodes. However, after the lateral branches were also pruned at certain points, more branches appeared from those regions. Later, when he mowed the lawn grass regularly by cutting the shoot tips, the grass appeared more bushy and dense within a few weeks.

Based on this observation, answer the following:

(a) Which type of meristematic tissue is responsible for the formation of new branches from the nodes after cutting? (1 mark)

(b) Explain why the grass plant becomes bushier after repeated cutting at the shoot tips. (1.5 marks)

(c) Name the process by which meristematic cells specialize into permanent tissues. (0.5 mark)

Section D: Short Answer Questions (5 marks)

6. Differentiate between simple permanent tissues and complex permanent tissues with one example each. (2 marks)

7. What is the main function of companion cells in phloem tissue? (1.5 marks)

8. Why are blood and bones both considered connective tissues despite their different properties? (1.5 marks)

Section E: Long Answer Questions (4 marks)

9. Describe the structure and functions of different types of joints in the human body. Explain how the structure of each joint type is related to its specific function. (4 marks)